Waves at boundaries

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe reflection and transmission at a boundary.

SUCCESS CRITERIA - I can...

1. Identify incident, reflected and transmitted waves.
2. Distinguish fixed-end from free-end pulse reflection.
3. Track energy without assuming equal amplitudes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “boundary” and “reflection” to explain your model or result. Sentence starters are optional.

boundary: the place where a medium or constraint changes

reflection: a wave returning from a boundary

transmission: a wave continuing through or beyond a boundary

01 Bring back an earlier idea

A reflected signal returns after 3 s.
Name the extra quantity needed to calculate distance and explain why this is a round trip.

Word help: signal = a disturbance that carries information

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $f_{reflected} = f_{incident}$.
At a stationary boundary. A displacement pulse returns inverted from an ideal fixed end and upright from an ideal free end.

Worked reasoning: At a fixed end, an upward rope pulse reflects downward. At an ideal free end it reflects without inversion. These are different boundary conditions.

Your turn: Draw incident and reflected pulse snapshots for both ends; show propagation directions.

Sentence starter: The … in my model represents … . Its label shows … .

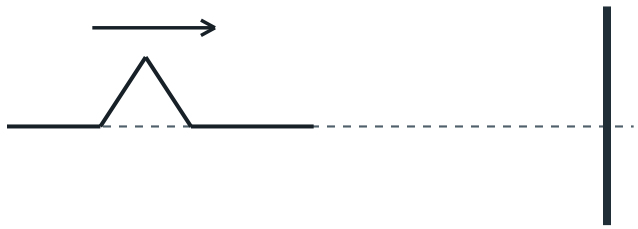
Marzano check - circle: ① ② ③

03 Look and explain

An upward rope pulse reaches an ideal fixed end.
Sketch the reflected pulse and its travel direction.

Word help: ideal = using stated simplifying conditions

Sentence starter: The … in my model represents … . Its label shows … .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Build the solution

A supplied energy split gives 10 J incident, 4 J reflected and no other loss.
Find transmitted energy.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Does a smaller transmitted amplitude prove frequency changed?
Explain what usually fixes the frequency at a stationary boundary.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg sample completely changes phase with latent heat 600 J/kg.
Find the energy magnitude required.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

For an ideal free-end rope reflection, is an upward displacement pulse returned upright or inverted?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two equal upward pulses approach each other on an ideal rope.
Sketch overlap and later separation.
Explain whether they permanently stick.

Word help: ideal = using stated simplifying conditions

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3